Eren Köselerli

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LinkedIn — Portfolio — GitHub

SUMMARY

Software Engineer with a shipped commercial track record in C# and C++. Specializing in scalable game architectures, custom engine development, and high-performance physics solvers. Proven expertise in engineering robust, loosely coupled codebases utilizing SOLID principles, Entity Component System (ECS) architectures, and strict memory optimization protocols.

EXPERIENCE

Blay Games July 2025 – March 2026
Game Developer

Project: KissCargo

- Engineered a low-latency real-time multiplayer synchronization system by integrating the Normcore (Normal Realtime) SDK, achieving sub-100ms replication of player states, dynamic inventory data, and physics-based grabbable objects across network clients.
- Architected a physics-based item throwing mechanic by applying the Strategy Pattern, enabling modular encapsulation of inventory, grabbed-object, and interactive throw behaviors with charge-up dynamics.
- Designed a highly modular and extensible Finite State Machine (FSM) AI framework utilizing ScriptableObjects, reducing the development cycle for new enemy behaviors by 50% across multiple distinct enemy types.

Project: OROBOROS

- Optimized runtime memory allocation and eliminated garbage collection (GC) frame spikes by refactoring over 10+ core controllers to use ZLinq (zero-allocation queries) and migrating the codebase to Unity 6's new physics APIs.
- Engineered the Steam Achievements integration for Standalone PC builds, as measured by implementing a dynamic sanitizer to convert local Quest IDs into Steam-compatible API keys and registering asynchronous event hooks to unlock achievements instantly upon quest completion.
- Eliminated animation desynchronization bugs and improved gameplay stability in boss encounters by refactoring legacy coroutine-based behavior into a modular, event-driven state machine synchronized with Unity Animation Events.

Blay Games October 2024 – March 2025
Game Developer Intern (Projects: Alice & GOKA)

- Prevented local game state tampering and memory hacking exploits, reducing unauthorized profile modification reports to 0% as measured by security audits, by integrating the CodeStage Anti-Cheat Toolkit, converting critical values to protected types, and implementing secure serialization.
- Enhanced gameplay balance and haptic feedback customizability, resulting in positive user feedback loops as measured by test play sessions, by updating core blocker/support controllers and refactoring vibration control states through EffectManager.

TESTINIUM July 2024 – August 2024
Software Test and QA Intern

- Ensured pre-release stability for high-scale applications as measured by comprehensive test coverage by executing strict quality assurance protocols and performance validations.

TECHNICAL PROJECTS

SimpleEngine2D / Tactical Tetris — *C++17, SFML 3*

- Designed and implemented a lightweight, high-performance 2D game engine from scratch, achieving complete decoupling of game logic from rendering and ensuring a locked 60 FPS frame rate by developing a custom Entity-Component-System (ECS) architecture in Modern C++ and SFML.
- Developed a fluid drag-and-drop block placement mechanic, achieving high-precision grid positioning and real-time collision previews by writing coordinate-translation algorithms that map screen-space mouse inputs to game board coordinates.

Slingshot — *Unity, C#*

- Engineered a high-precision, custom 2D Kinematic Physics Solver and a modular Player Finite State Machine (FSM), achieving 100% deterministic platformer controls and eliminating wall-clipping bugs.
- Optimized memory allocations by replacing traditional Unity coroutines with an asynchronous action engine utilizing C# UniTask and polymorphic serialization, allowing native configuration of complex physics events.

Into the Night — *Unity, C#*

- Accomplished a modular and highly scalable turn-based combat system, reducing state-transition bugs by 40%, by implementing the State Design Pattern to isolate 6 distinct gameplay phases into decoupled, independent execution blocks.
- Developed a highly reusable Generic Factory Pattern, reducing scene asset file size and memory footprint by 40%, by centralizing the instantiation of players, enemies, and environments using ScriptableObject databases.

EDUCATION

Izmir University of Economics 2020 – 2025
Bachelor of Software Engineering

TECHNICAL PROFICIENCIES

- **Languages:** C#, C++, Java, Python
- **Engines & Platforms:** Unity, Godot, GDExtension
- **Architecture & Systems:** DOTS, Normcore Networking, Design Patterns, Data Structures, 2D/3D Physics
- **Workflows & Tools:** Git, Agile Methodologies, Software QA, Odin Inspector, Blender